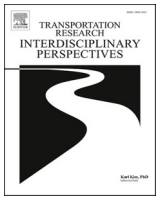
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Variety seeking route choice behavior of commuting e-cyclists retrieved from GPS data

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ABSTRACT

The rapid adoption of e-bicycles challenges traditional route choice insights, which typically assume that cyclists minimize physical effort and adhere to habitual paths. By partially decoupling travel distance from physical exertion, e-bicycles may relax efficiency-oriented constraints and enable greater flexibility in route choice. This study introduces the concept of Variety Seeking Route Choice Behavior (VSRCB) to quantify the extent to which cyclists deviate from habitual routes, and examines whether electric assistance enhances this flexibility. Using high-resolution GPS trajectory data, we develop a multi-dimensional framework to quantify route variety among Dutch cyclists. The results confirm that VSRCB is an observable phenomenon across both modes, though e-cyclists exhibit higher levels than traditional cyclists. A decomposition analysis indicates that this variety primarily arises from more frequent switching among available route alternatives (Quantity and Balance), rather than from substantial geometric deviation (Distinction). Moreover, VSRCB is most pronounced in highly urbanized areas, where dense road networks support flexible routing, while demographic characteristics show no significant association. These findings suggest that cycling infrastructure planning should move beyond single high-capacity corridors and instead emphasize redundant, mesh-like networks that better accommodate the flexible routing behavior of e-cyclists.

Introduction

The electrification of micromobility has altered the landscape of urban transportation. In 2023, approximately 29% of all bicycles sold in the European Union were electric, representing a substantial increase from just 6% in 2015 (Statista, 2024). Unlike traditional bicycles, e-bicycles provide pedal assistance that significantly reduces physical exertion, enabling riders to travel longer distances at higher speeds (Van Cauwenberg et al., 2018; Jenkins et al., 2022). While the environmental and health benefits of this shift are well-documented (Casier and Witlox, 2022; Elliot et al., 2018; Schneider, 2023), the impact of electrification on the spatial structure of route choice behavior remains under-explored.

Traditionally, route choice approaches for cyclists are often predicated on a utility-maximization framework, where cyclists are assumed to minimize disutility, primarily defined by physical effort, travel time, and distance (Fitch and Handy, 2020; Łukawska, 2024; Meister et al., 2023). For traditional cyclists, the physical exertion required to propel the vehicle creates a high impedance to deviation; taking a longer or different route incurs a tangible physiological cost. Consequently, traditional commuting routes are typically characterized by high stability and repetition, as riders converge on the single most efficient path to minimize energy expenditure (Broach et al., 2012; Heinen et al., 2011; Ton et al., 2017).

However, the advent of e-bicycles challenges this established paradigm. By providing electric assistance, e-bicycles decouple travel distance from physical exhaustion (De Kruijf et al., 2018). This technological shift significantly lowers the marginal physical cost of detouring. While arrival time remains a binding constraint for commuters (Broach et al., 2012; Łukawska, 2024), this technological shift creates a theoretical possibility: the reduced physical constraint could empower cyclists to trade marginal time differences for richer riding experiences.

To investigate this potential behavioral shift, we introduce the concept of Variety Seeking Route Choice Behavior (VSRCB). Variety Seeking Behavior (VSB) is well-established in marketing as a drive to

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alternate between options to satisfy curiosity (Kahn, 1995; McAlister and Pessemier, 1982; Ratner and Kahn, 2002; Zhang, 2022). Commuting is inherently a high-frequency (Bernardi et al., 2018), repetitive activity that can induce boredom. This raises a fundamental question: In the context of daily commuting, do cyclists actively seek variety to break this monotony, or do they strictly adhere to a single, most efficient route? While the desire for variety may be intrinsic, its manifestation in cycling is likely constrained by physical effort. For traditional cyclists, the physiological cost of deviating from the optimal path constitutes a tangible barrier to exploration. Theoretically, electric bicycles mitigate the constraint of physical fatigue, potentially enabling cyclists to be liberated from the constraints of the shortest path to seek novelty, enjoy scenic variations, or break the monotony of repetitive commuting (Park and Akar, 2019; Plazier et al., 2017; Standen et al., 2017; Vedel et al., 2017). This further leads to the question: Does the reduction of physical effort allow the latent demand for variety to manifest, or does the habit of repetition dominate regardless of the mode?

Despite the potential for changed behavior, existing research on e-cyclists largely mirrors that of traditional cyclists, focusing heavily on static environmental factors such as slope gradients or infrastructure quality (Chavis and Martinez, 2021; Hardinghaus and Weschke, 2023; Khavarian et al., 2024; Meister et al., 2023; Plazier et al., 2017). While these approaches explain why cyclists choose a specific route based on route characteristics (preference heterogeneity), they do not investigate whether cyclists alternate between different routes for the same travel purpose (e.g., commuting) over time. Consequently, there is a lack of empirical research quantifying actual route variety seeking behavior in travel behavior. It remains unclear whether spatial route diversity is a genuine behavioral phenomenon in commuting or merely a theoretical concept, and to what extent physical assistance influences this behavior.

Understanding this behavioral shift is crucial for two reasons. First, from a behavioral science perspective, it tests whether "spatial emancipation" translates into actual behavioral change. In doing so, the study explores VSRCB as a personal characteristic that may contribute to the decision-making process—aligning with intrinsic motivations similar to 'openness to experiences' (O'Hern et al., 2020) —thereby addressing the growing need for detailed segmentation of cyclist groups (Lilasathapornkit et al., 2025). Second, and more importantly for policy, it challenges the traditional approach to infrastructure planning. If e-cyclists are proven to be "variety seekers" who are willing to utilize a mesh of alternative routes, urban planning can move beyond building single, high-capacity cycle superhighways toward developing resilient, grid-like networks that distribute traffic flow and prevent congestion. These insights are important for investment decisions (Cunha and Silva, 2023; Meng, 2022; Van Nijen et al., 2024; Zhao and Manaugh, 2023) and promotional activities (Fields et al., 2022; Khavarian et al., 2024).

To address these gaps, this study employs high-resolution GPS trajectory data to objectively quantify and analyze the variety seeking route choice behavior (VSRCB) of e-cyclists. Unlike stated preference (SP) surveys, which are susceptible to hypothetical bias (Hensher, 2010), GPS data capture behavior under real-world conditions. While traditional revealed preference (RP) surveys also reflect actual behavior, they are often labor-intensive to collect and limited in scale due to the need for detailed manual reporting (Peer et al., 2013). In contrast, GPS data provide high-resolution, continuous, and precise tracking of individual movements, offering detailed information on actual route choices over time (Hess et al., 2015). This approach overcomes the limitations of both SP and traditional RP methods by reducing inaccuracies associated with self-reported data and ensuring a higher quality of spatial information. Specifically, this study aims to:

(i) develop a multi-dimensional measurement framework for VSRCB that characterizes route use in terms of the number of routes taken, the balance of their utilization, and their spatial differentiation;
(ii) compare the VSRCB of e-cyclists with that of traditional cyclists, thereby assessing whether electric assistance is associated with greater route diversity through the relaxation of physical constraints;
(iii) investigate the relationship between VSRCB and cyclist profiles, examining whether observed differences in route variety are more closely associated with individual demographic characteristics (e.g., age, gender) or shaped by the spatial context of the home location (e.g., urbanization levels).

The remaining sections of this paper are structured as follows. Following this introduction, Section 2 outlines the methodology employed in the construction of the VSRCB measurement. Section 3 provides a detailed account of data processing methods employed. Section 4 presents the results of the VSRCB and an analysis of the factors that influence it. Finally, Section 5 presents a summary of the findings, policy recommendations based on the results, an analysis of the limitations of this study, and suggestions for future research.

Methodology

The VSRCB index

To fulfill the research objectives outlined in the introduction and to empirically assess the route choice patterns of cyclists, this study establishes a quantitative measurement framework for VSRCB. Unlike traditional metrics that often rely on single-dimensional indicators, this framework conceptualizes route choice variety as a comprehensive, multidimensional construct designed to capture the complexity of commuting behavior. Accordingly, the VSRCB index is decomposed into three complementary dimensions: Quantity, Balance, and Distinction. These dimensions collectively measure the size of the route set, the evenness of usage distribution, and the spatial uniqueness of the chosen paths, respectively.

First, the Quantity dimension captures the extent of route exploration by measuring the number of distinct routes chosen by a respondent for a given OD pair. Holding other factors constant, a larger set of utilized routes reflects a higher degree of route choice variety.

Second, the Balance dimension reflects the evenness of route usage across the available alternatives. Rather than focusing solely on how many routes are chosen, this dimension evaluates how frequently each route is selected over time. A more balanced usage pattern—where routes are chosen with similar frequencies—indicates stronger variety-seeking behavior than a highly concentrated pattern dominated by a single route.

Third, the Distinction dimension accounts for the spatial uniqueness of the selected routes. Even when multiple routes are used, their contribution to route choice variety depends on the degree to which they are geometrically distinct. Routes with less physical overlap and greater spatial divergence contribute more positively to the overall VSRCB, as they represent genuinely different travel experiences.

Based on this multidimensional conceptualization, we define the VSRCB index at the level of a single directional OD pair for a specific respondent. Consequently, distinct trip chains (e.g., return trips or trips with different OD pairs) are evaluated as independent analysis units to ensure methodological consistency.

Formally, the VSRCB index integrates the three dimensions—Quantity, Balance, and Distinction—according to the following formulation:

$$VSRCB = \sum_{i \in K} w_i \times DR_i \times URI_i \tag{1}$$

where K denotes the set of all unique routes chosen by a respondent for the specified OD pair. The summation over K operationalizes the Quantity dimension by aggregating the contribution of each distinct route to the overall index.

The term w_i represents the entropic weight associated with route i, capturing the Balance dimension. By assigning weights based on the entropy contribution of each route, this component rewards a more uniform distribution of route selection frequencies.

The Distinction dimension is jointly quantified by two spatial metrics. The Distinction Rate (DR_i) measures the geometric uniqueness of route i based on its spatial overlap with other routes in the choice set, while the Unique Route Index (URI_i) further refines this measure by accounting for the magnitude of spatial deviation among independent routes. Together, these two terms ensure that routes contributing greater spatial diversity exert a stronger influence on the VSRCB value.

The following sections provide a detailed description of the mathematical formulation and computational procedures for each component of the VSRCB index.

Mathematical formulation of VSRCB dimensions

As defined in Eq. (1), the VSRCB index aggregates the contribution of each selected route. While the Quantity dimension is inherently captured by the summation over the route set K, the Balance and Distinction dimensions require specific mathematical formulations to quantify the weight and spatial uniqueness of each route. This section details the derivation of these components.

Balance: entropic weighting

To accurately reflect the Balance dimension within the VSRCB index, this measurement employs the concept of entropy as defined within the field of information theory (Shannon, 1948). A key principle of entropy is that a uniform distribution of probabilities maximizes the value, representing the highest level of unpredictability or diversity. In the context of route choice, if all routes are chosen with equal frequency, entropy reaches its maximum, suggesting a perfectly balanced distribution of route selections. This uniformity implies no single route is predominantly favored, indicating a high level of variety seeking behavior.

Following this principle, the entropic weight associated with route i, denoted as w_i is defined as:

$$w_i = -p_i \log_2(p_i) \tag{2}$$

where p_i represents the probability that route i is chosen. In this study, the logarithm is computed using base 2 ($\log_2$), such that the entropy contribution is expressed in bits.

The selection probability p_i is derived from observed route choice frequencies and is calculated as:

$$p_i = \frac{f_i}{\sum_{j \in K} f_j} \tag{3}$$

where f_i denotes the number of times a respondent selects route i, and K represents the set of all unique routes chosen by the respondent for the specified OD pair.

Distinction: spatial overlap

While the entropic weights established in the previous section capture the frequency-based balance of route usage, they do not account for the spatial geometry of the paths. To operationalize the Distinction dimension, it is necessary to quantify the degree of spatial uniqueness of each route.

First, we assess the geometric distinctiveness between any specific pair of routes. The pairwise distinction rate, denoted as $DR_{i,j}$, measures the extent to which route i differs spatially from another route j selected by the same respondent. It is calculated as follows:

$$DR_{i,j} = 1 - \frac{SL_{i,j}}{L_i} \tag{4}$$

where $SL_{i,j}$ represents the length of shared links (geometric overlap) between route i and route j, and L_i is the total length of route i. This calculation quantifies the distinctiveness of route i relative to route j by determining the proportion of the route's length that is non-overlapping. A value of 1 indicates that the routes are completely spatially disjoint, while a value approaching 0 indicates high redundancy.

To derive a single distinctiveness value for route i within the context of the entire choice set, we calculate the average of its pairwise distinction rates against all other routes considered by the respondent. The overall distinction rate (DR_i) is defined as:

$$DR_i = \frac{\sum_{j \in K, j \neq i} DR_{i,j}}{N - 1} \tag{5}$$

where N is the total number of distinct routes selected by the respondent (i.e., the cardinality of set K).

Distinction: scale of independence

A critical limitation of the overlap-based metric (DR) is its insensitivity to the length of completely independent routes. An independent route is formally defined as a route that has no overlapping segments with any other routes within the respondent's set of commuting routes. By definition, if a route shares no common links with others in the choice set ($SL = 0$), its distinction rate is mathematically fixed at unity ($DR = 1$).

This creates a limitation where the DR metric assigns an identical distinctiveness score to all independent routes, regardless of their length differences. Consequently, a very short independent path contributes the same value to the index as a significantly longer one, failing to account for the difference in travel distance.

To address this and refine the Distinction dimension, we introduce the Unique Route Index (URI). This component scales the contribution of independent routes based on their length relative to the respondent's average route length ($\overline{L}$). The URI is calculated as follows:

$$URI_i = \begin{cases} \log_2\left(1 + \dfrac{L_i}{\overline{L}}\right), & \text{if Route } i \text{ is an independent route} \\ 1, & \text{if Route } i \text{ is not an independent route} \end{cases} \tag{6}$$

where L_i is the length of route i, and $\overline{L}$ is the average length of all routes selected by the respondent, calculated as $\overline{L} = \sum_{i \in R} L_i / N$. The logarithmic transformation is employed to dampen the effect of extremely long paths, ensuring that the index rewards genuine exploration without being disproportionately skewed by outliers.

Numerical example

To illustrate the comprehensive calculation process of the VSRCB index, this section applies the derived mathematical formulas to a hypothetical route choice scenario. The example demonstrates how the dimensions of Quantity, Balance, and Distinction are synthesized into a single metric using the general formula in Eq. (1).

The analysis is based on the schematic road network shown in Fig. 1. The scope is strictly limited to the directional OD pair from Home (O) to Work (D). Within this network, four valid routes (A, B, C, and D) are identified. Table 1 provides the detailed specifications of the network. It lists the length of each individual link and defines the topological composition of each route.

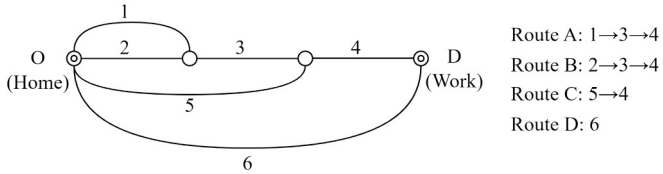


Fig. 1. Road network diagram.

Table 1
Characteristics of network links and route composition.

Link ID	Link attributes	Route composition			
	Length	Route A	Route B	Route C	Route D
1	12	√			
2	10		√		
3	10	√	√		
4	10	√	√	√	
5	25			√	
6	50				√
Total length	–-	32	30	35	50

We define two hypothetical respondents to represent different route choice behaviors. Respondent 01 is characterized by a choice set of three overlapping routes (Routes A, B, and C), whereas Respondent 02 expands this selection to include all four available routes (Routes A, B, C, and D), thereby incorporating the spatially independent path. For this illustrative example, we assume a baseline scenario where each chosen route is selected with equal frequency. This allows for a clear, step-by-step verification of the computational procedure.

Table 2 details the step-by-step calculation results for both respondents. This process begins by calculating the entropic weight (w_i) for each route based on selection frequency. Subsequently, the spatial distinctiveness is assessed through the distinction rate (DR_i) and, where applicable, refined by the Unique Route Index (URI_i).

As shown in Table 2, Respondent 02 achieves a significantly higher total VSRCB score (1.701) compared to Respondent 01 (0.922). This difference is driven by two key factors verified by the calculation.

Quantity Dimension: Respondent 02 utilizes a larger set of routes ($N = 4$ vs. $N = 3$), which fundamentally expands the basis for variety seeking.

Distinction Dimension (Role of Independent Routes): The inclusion of Route D by Respondent 02 plays a critical role. Route D is a completely independent route with no spatial overlap ($SL_i = 0$), resulting in a raw distinction rate (DR_i) of 1.0. However, unlike the overlap-based *DR* metric, the *URI* further refines Route D's contribution. As shown in Table 2, the *URI* for Route D is calculated as 1.239.

This example confirms that the VSRCB index functions as a comprehensive metric: it rewards not only the number of routes chosen and the balance of their usage, but also accurately weighs the spatial uniqueness of independent paths through the combined application of *DR* and *URI*.

Data processing

According to the methodology outlined, it is essential to transform the raw GPS data into route data. The entire data processing procedure consists of the following steps.

Step 1: Data Collection. This initial stage involves the integration of various data sources, including GPS trajectory data, travel pattern data, and geographic data.

Step 2: GPS Data Pre-Processing. The raw GPS data undergoes preliminary filtering to remove any outliers or anomalies. This stage also includes data matching and cleaning to ensure that only relevant and reliable travel pattern data are retained for analysis.

Step 3: Map Matching. In this step, the cleaned GPS data are aligned with the existing road network using a Hidden Markov Model (HMM) developed by Meert and Verbeke (2018).

Step 4: Route Set Filtering. The final step focuses on refining the dataset by eliminating erroneous routes and retaining only those respondents who commute at least twice, thus producing a robust dataset for further analysis.

Data collection

This study integrates GPS trajectory data, travel pattern data, and geographic data, to observe the phenomenon of variety seeking behavior in route choice, with a primary focus on e-cyclists in the Dutch province of Noord-Brabant and its neighboring regions. Traditional cyclists are also included in the analysis to serve as a comparative benchmark.

GPS trajectory data originates from the "B-Riders" bicycle stimulation program in Noord-Brabant. Initiated in 2013, the B-Riders project aimed to facilitate a modal shift from cars to cycling by offering financial rewards based on kilometers ridden, supplemented by personal coaching (De Kruijf et al., 2018). Participants used a dedicated mobile application to register their trips, enabling the continuous recording of GPS traces. In addition to dynamic trajectory data, the program also collected static sociodemographic information from participants upon registration, including age, gender, and the 6-digit postal codes of their home and work locations. While the project initially targeted e-cyclists (Phase 1), it subsequently expanded to welcome all cyclists in its second phase. This study utilizes data recorded via participants' smartphones from January to September 2014, with participation periods varying among different respondents. Consequently, while the dataset is rich in e-cycling trajectories, it also systematically captures traditional cycling trips.

These raw traces were subsequently processed to infer specific transportation modes (e.g., car, e-bicycle, traditional bicycle) and organized into structured segments (Feng and Timmermans, 2019). From the inferred dataset, we specifically extracted home-to-work commuting trajectories identified as e-bicycle trips for the main analysis, and traditional bicycle trips for comparative purposes. The data set provides a detailed account of each respondent's trajectory, including identifiers, transportation modes, timestamps, spatial coordinates, postal codes, and trip purposes. Using the trip purpose attributes, only trips originating from 'Home' and ending at 'Paid Work' were retained. A sample of the processed travel pattern data structure is presented in Table 3.

The geographical scope of this study focuses on the province of Noord-Brabant, situated in the southern part of the Netherlands, as illustrated in Fig. 2a. To construct an accurate spatial reference, this study employs the Dutch National Road Network (Nationaal Wegenbestand, or NWB) Shapefile. To align with the temporal coverage of the GPS trajectories, the specific version corresponding to the 2014 road network was utilized. As shown in Fig. 2b, this dataset provides a coherent and detailed transportation network that covers not only

Table 2
Calculation VSRCB for Respondents 01 and 02.

Respondent 01	w_i	$DR_{i,j}$			DR_i		URI_i	$VSRCB_i$	VSRCB
		A	B	C					
A	0.528		0.375	0.688	0.531		1	0.281	0.922
B	0.528	0.333		0.667	0.500		1	0.264	
C	0.528	0.714	0.714		0.714		1	0.377	
Respondent 02	w_i	$DR_{i,j}$				DR_i	URI_i	$VSRCB_i$	VSRCB
		A	B	C	D				
A	0.500		0.375	0.688	1.000	0.688	1	0.344	1.701
B	0.500	0.333		0.667	1.000	0.667	1	0.333	
C	0.500	0.714	0.714		1.000	0.810	1	0.405	
D	0.500	1.000	1.000	1.000		1.000	1.239	0.620	

Table 3
Sample of processed travel pattern data (filtered for home-to-work commuting).

User ID	Mode	Origin				Destination			
		Timestamp	Lat	Lon	PC	Timestamp	Lat	Lon	PC
2397	BIKE	2014/1/8 9:07:54	51.60395	4.78852	4826HG	2014/1/8 9:16:18	51.58947	4.77549	4811KC
2397	BIKE	2014/1/10 7:02:26	51.60214	4.78555	4826HG	2014/1/10 7:07:52	51.59532	4.77881	4811KC
2397	EBIKE	2014/1/21 7:37:34	51.60260	4.78963	4826HG	2014/1/21 7:48:47	51.58916	4.78348	4811KC
…	…	…	…	…	…	…	…	…	…

Note: 'PC' denotes the 6-digit Dutch postal code. 'Mode' refers to the transportation mode inferred from GPS trajectories.

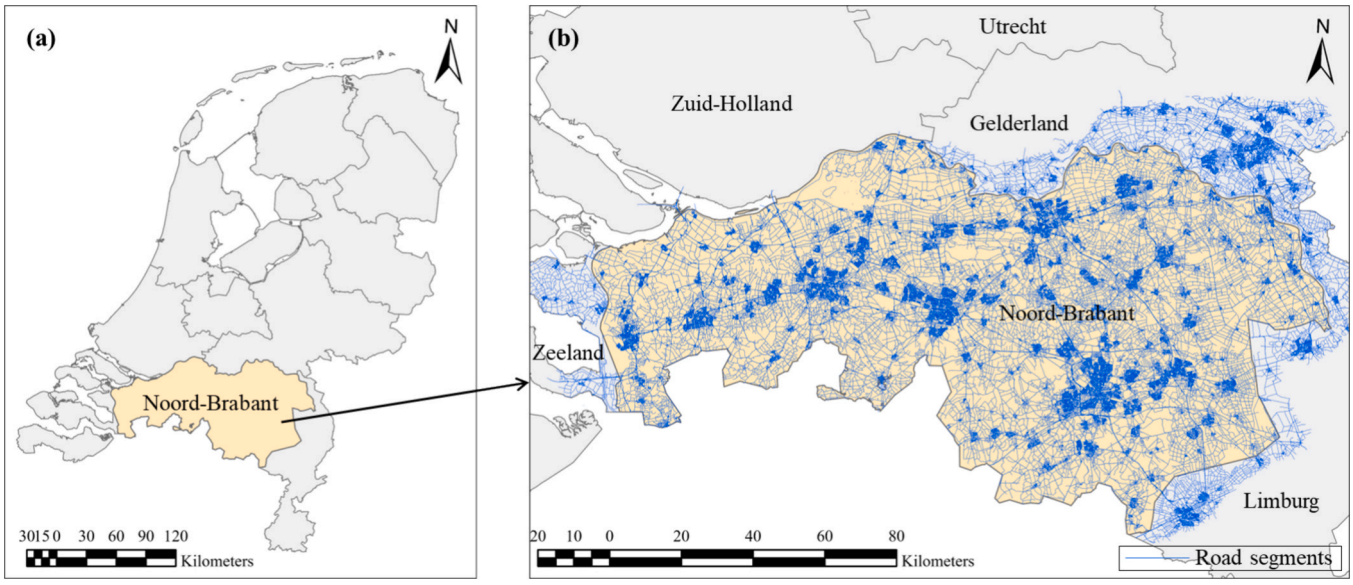


Fig. 2. Overview of the study area and road network. *Note*: The NWB data is publicly available at https://downloads.rijkswaterstaatdata.nl/nwb-wegen/geogegevens/shapefile/.

Noord-Brabant but also extends into adjacent provinces, including Gelderland, Limburg, Zuid-Holland, and Zeeland.

Step 2: GPS data Pre-Processing

Following the data collection and initial selection described in Section 3.1, the raw GPS trajectories require rigorous pre-processing to ensure the accuracy of the subsequent route choice analysis (Lißner and Huber, 2021). To ensure the validity of the comparative analysis, identical pre-processing protocols were applied to both the primary e-cyclist dataset and the traditional cyclist benchmark group. This stage primarily involves data matching, OD alignment, and noise removal.

First, the extracted travel pattern segments were matched with the raw GPS repository to reconstruct the complete spatial trajectory. While the travel pattern data defines the boundaries of a trip (i.e., origin, destination, and duration), it lacks the intermediate path details essential for the VSRCB analysis. To address this, we employed a timestamp-based matching approach. For each identified commuting segment, all raw GPS points recorded by the same respondent within the specific start and end timestamps were retrieved and aggregated. This method was selected for its superior reliability over purely geographic matching. Geographic approaches often introduce ambiguities, particularly at journey endpoints where stationary waiting times (dwell time) generate spatial clusters that can skew trajectory inference. By utilizing precise timestamp alignment, we successfully extracted the continuous GPS traces corresponding to each discrete commuting trip.

Following the retrieval of raw trajectories, a critical step was to ensure the spatial consistency of OD locations. In raw GPS datasets, the recorded coordinates for "Home" or "Work" for the same respondent frequently exhibit spatial variance across different days due to signal fluctuations and minor behavioral differences (e.g., parking at different spots). This inconsistency poses a challenge for comparative route analysis, which requires a fixed OD pair for each respondent. To address this, we implemented a two-stage process: OD verification and spatial alignment.

First, a rigorous quality control procedure was applied based on the postal code areas registered by respondents. Raw postal code data were repaired (e.g., formatting corrections) to serve as a reliable spatial reference. A trip was retained only if its GPS start point fell within the respondent's home postal code area and its end point within the work postal code area. This filtering step effectively removed non-commuting trips and erroneous GPS records.

Subsequently, to unify the spatially scattered OD points into a consistent pair for each respondent, we evaluated two alignment methods: (1) The Centroid Replacement Method, which substitutes OD points with the geometric centroid of the respective postal code area; and (2) The Average Coordinate Method, which calculates the mean latitude and longitude of all valid start and end points for a given respondent.

Fig. 3 illustrates the comparative evaluation of these two methods using respondent 2962 as an example. As shown, the centroid of the postal code area is significantly displaced from the actual cluster of GPS start points, introducing a substantial spatial error that could severely compromise subsequent map matching. In contrast, the average coordinate is situated within the dense cluster of actual GPS recordings, minimizing deviation from the true travel path. Consequently, the Average Coordinate Method was adopted to replace the start and end points of all trajectories, ensuring both spatial consistency and high fidelity to the original data.

Finally, the spatially aligned trajectories underwent a rigorous

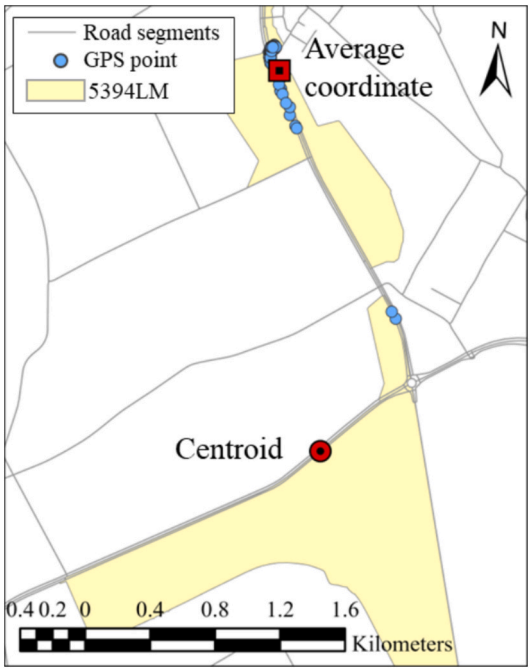


Fig. 3. Comparison of OD alignment methods.

cleaning process to eliminate signal noise and kinematic anomalies (Gao et al., 2024). Raw GPS data frequently contain errors that manifest as drift points or velocity outliers, which can severely disrupt the topological integrity required for accurate map matching.

The primary focus was the elimination of drift points. As illustrated in Fig. 4, signal multipath errors often create "burrs" in the trajectory—sharp, unrealistic deviations from the smooth travel path. These deviations form acute angles between consecutive track segments. To rectify this, we implemented an iterative angle-based filtering algorithm comprising the following steps:

1. Chronological Sorting. All GPS points within a trip were indexed strictly by timestamp.
2. Angle Calculation. The angle formed by any three consecutive GPS points (S_{i-1}, S_i, S_{i+1}) was computed.
3. Threshold Filtering. A point S_i was classified as a drift point if the calculated angle was less than a predefined threshold.

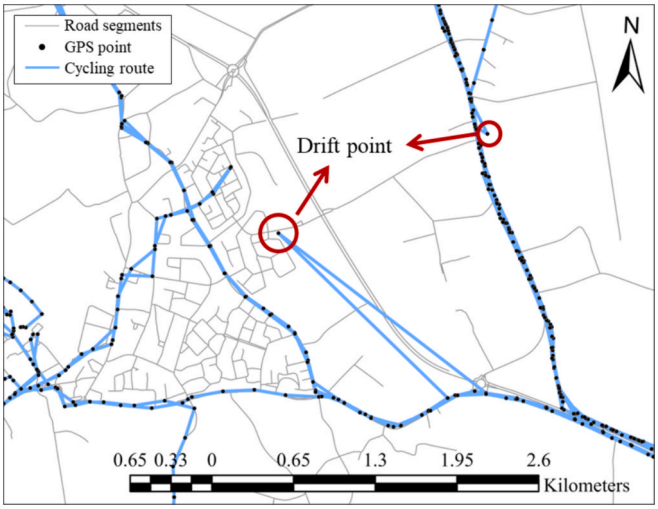


Fig. 4. Illustration of GPS drift points.

4. Iterative Removal. Identified drift points were removed, and the process was repeated iteratively on the remaining sequence until all consecutive segments exhibited angles above the threshold.

This iterative smoothing process effectively removed the sharp "burrs" shown in Fig. 4, ensuring the geometric continuity of the route.

In parallel, velocity anomalies were addressed. Instantaneous speeds were calculated between consecutive points. Segments implying speeds exceeding the physical capabilities of e-cycling (or traditional cycling, respectively) were flagged as errors and removed. This comprehensive pre-processing provides a clean, robust dataset, establishing a solid foundation for the subsequent map-matching phase.

Step 3: Map matching

Map matching is pivotal for aligning discrete GPS points with the underlying topological road network to reconstruct continuous travel paths. This study employs a Hidden Markov Model (HMM) with Non-Emitting States, implemented via the Leuven.MapMatching library (Meert and Verbeke, 2018). Unlike simple nearest-neighbor projection, the HMM approach considers the connectivity of the entire sequence, determining the most probable path by balancing observation probability (GPS proximity) and transition probability (network topology).

Prior to matching, the NWB road network (refer to Fig. 2b) underwent topological processing. Node and link attributes were extracted, and bidirectional graph connectivity was established by duplicating geometries for two-way segments. This ensured a traversable graph structure for the HMM algorithm.

Upon the completion of the HMM calculation, a validation procedure was implemented to verify the success rate. Since manually inspecting thousands of routes is unfeasible, we established a distance-based automated filter to identify incomplete or erroneous matches. The logic of this filter is twofold:

1. Connectivity Check. The algorithm checks if the matched route forms a continuous path from origin to destination without topological breaks.
2. Endpoint Distance Check. We calculated the Euclidean distance between the final GPS point of a trip and the last node of the matched road segment. In this study, the search radius threshold was set to 100 m. While previous research suggests that a 50-meter radius is generally sufficient to cover GPS error and capture true road segments (Gao et al., 2024), we adopted a slightly relaxed threshold of 100 m. This adjustment was made to accommodate the operational characteristics of e-bicycles and to ensure robustness following the averaging of endpoint coordinates. Consequently, if the calculated distance exceeded this 100-meter threshold, it indicated that the HMM failed to find a feasible path to the true destination, classifying the result as an incomplete match.

Routes flagged by this filter were subsequently examined visually to diagnose the causes. As illustrated in Fig. 5, these anomalies primarily stemmed from two issues:

1. Network Discontinuities (Fig. 5a, 5b). Interruptions in the digital road network (e.g., missing links or unconnected nodes in the NWB shapefile) prevented the HMM from finding a valid transition between consecutive GPS points. This often occurs where cyclists utilize informal paths or shortcuts not recorded in the official road database.
2. Sparse Sampling (Fig. 5c). In instances where GPS sampling intervals were large (resulting in missing intermediate points), the algorithm faced ambiguity in inferring the specific path through complex urban intersections.

Based on the filtered dataset from Section 3.2 (Total 9,558 trips,

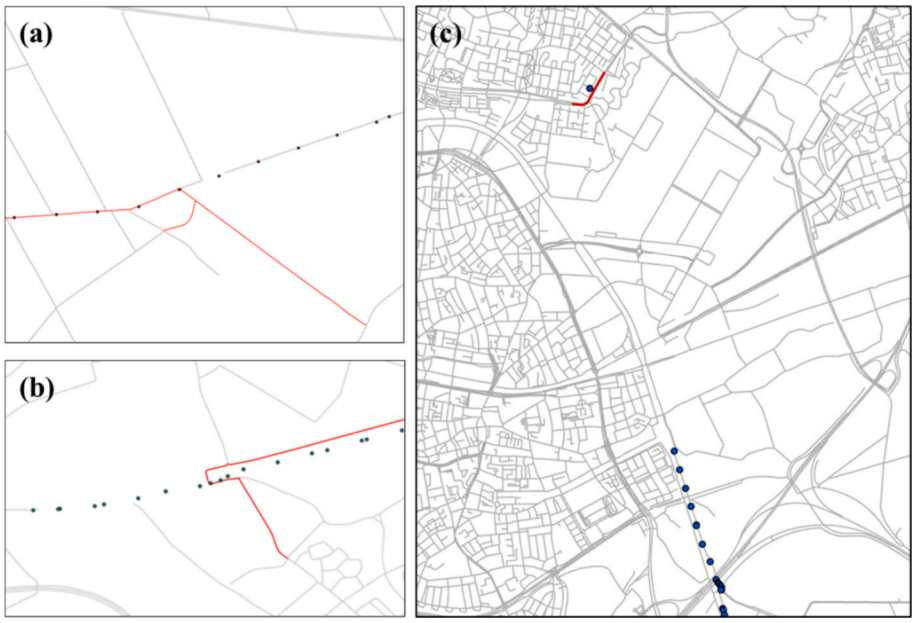


Fig. 5. Examples of map matching anomalies.

comprising both e-cyclist and traditional cyclist trajectories), the validation process identified 612 anomalous routes, representing 6.4% of the data. These unreliable segments were excluded to prevent bias in the route choice analysis. The final dataset retained 8,946 high-fidelity routes, providing a robust foundation for the subsequent analysis. Fig. 6 presents an example of a successfully matched home-to-work commuting route set for respondent 3163.

Step 4: Route set filtering

Following the map matching and validation process, the dataset underwent a final behavioral filtering stage to ensure the suitability of the routes for variety seeking analysis. This process involved two specific criteria applied uniformly to all respondents.

First, a length-based threshold was applied to exclude non-commuting detours. For each respondent, a route set was constructed containing all their recorded trajectories. Individual routes within this set were flagged as anomalies if their length exceeded twice the length of the shortest commute route recorded for that individual. This conservative threshold is informed by the findings of Chung et al. (2024), who report an average circuity factor of approximately 1.56 for cyclists. Given the reduced physical effort associated with e-cycling, a higher tolerance for route deviation is reasonable. In our dataset, applying a threshold of 2.0 resulted in the exclusion of only three extreme outliers, thereby retaining genuine variety seeking detours while filtering out routes unlikely to represent commuting behavior. After removing these outliers, the total number of valid routes was refined to 8,943.

Second, a frequency-based filter was implemented. To effectively quantify variety seeking behavior (VSRCB), longitudinal data is essential; variability cannot be discerned from a single trip. Therefore,

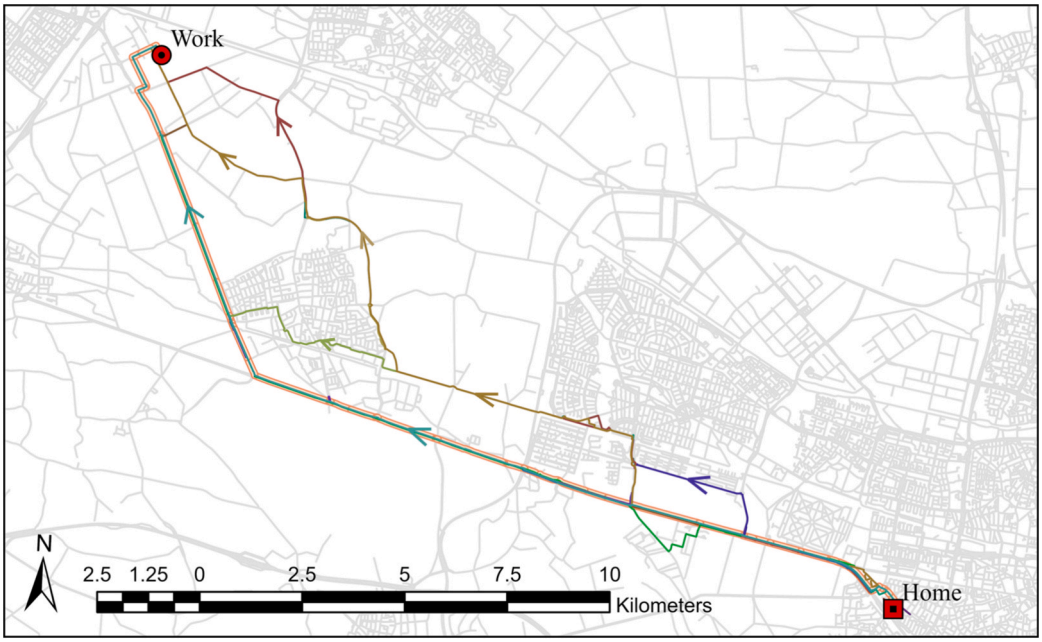


Fig. 6. Home-to-work commuting routes of Respondent 3163.

respondents with fewer than two recorded commuting trips were excluded from the analysis.

As a result of this rigorous filtering process, the final dataset was stratified into two distinct groups to facilitate the study's comparative analysis:

1. Primary Group (E-Cyclists): A total of 666 respondents were identified as commuting e-cyclists, serving as the main subject of this study.
2. Benchmark Group (Traditional Cyclists): A subset of 67 respondents using traditional bicycles was retained to serve as a comparative control group.

Notably, this stratification reveals a subset of 54 "dual-mode" respondents who are present in both groups. These individuals utilized both e-bicycles and traditional bicycles for commuting during the study period, providing a unique opportunity for within-subject analysis to control for personal heterogeneity. The final consolidated dataset comprises 8,655 high-fidelity commuting routes, ready for the calculation of VSRCB indicators.

Results and discussions

Personal characteristics and behaviors

Table 4 presents the personal characteristics of both e-cyclists and traditional cyclists in this study, contrasting them with representative data from the Dutch National Travel Survey (CBS, 2015). The urbanization data presented is derived from Statistics Netherlands, which classifies urbanization into five levels based on address density. In this study, these are merged into three categories: Level I (Highly Urbanized, >1,500 addresses/km^2), Level II (Moderately Urbanized, 1,000–1,500 addresses/km^2), and Level III (Lightly Urbanized, <1,000 addresses/km^2). Compared to the OViN benchmark, the study sample exhibits specific demographic deviations, notably a higher average age and a concentration in less urbanized areas. While the generalizability of the outcomes is limited, this does not pose a concern for the current study, which is primarily focused on developing a new measurement method for variety seeking behavior in route choices.

Fig. 7 compares the monthly commuting dynamics of e-cyclists (a) and traditional cyclists (b). The bar charts (left axis) display the total trip volume and the number of active commuters, while the line charts (right axis) represent the average commuting frequency per respondent. A clear seasonal trend is observed for both groups: the total number of commuters and trips declines notably during the summer months (July and August), reflecting the Dutch summer vacation period. However, a comparative analysis reveals a distinct difference in usage intensity. Throughout the observation period, e-cyclists consistently exhibit a higher average commuting frequency compared to traditional cyclists. Specifically, while the average frequency for cyclists generally fluctuates between 1.5 and 2.5 trips per month, e-cyclists maintain a baseline above 2.5, peaking at nearly 4.0 in January.

Table 4
Distribution of personal characteristics in our study (E-cyclists and Traditional cyclists) as compared to the representative Dutch National Travel Survey.

Attribute	Category	E-cyclists (N = 666)	Traditional cyclist (N = 67)	OviN (N = 8093)
Gender	Female	51%	45%	56%
	Male	49%	55%	44%
Age	≤ 45	24%	24%	39%
	(45,55]	47%	37%	20%
	> 55	29%	39%	41%
	Avg.	50.6	52.3	49.7
Home location urbanization level	I	17%	18%	46%
	II	21%	18%	17%
	III	62%	64%	37%

Fig. 8 compares the spatial characteristics of the commuting routes utilized by the two groups. Fig. 8a presents the distribution of route lengths. The statistical analysis indicates that e-cyclists generally engage in longer commutes compared to conventional cyclists. Specifically, the e-cyclist group exhibits a mean route length of 21.81 km (SD = 8.77), whereas the traditional cyclist group averages 20.20 km (SD = 9.51). Although the distributions overlap in the shorter distance range, the e-cyclist data shows a stronger tendency towards the longer-distance spectrum.

Regarding route diversity, Fig. 8b reveals a behavioral divergence. The histogram displays the distribution of the number of distinct routes chosen per respondent. Traditional cyclists exhibit a relatively conservative spatial pattern, relying on a limited set of path options (Mean = 4.48, SD = 4.79; Median = 3). In sharp contrast, e-cyclists demonstrate substantial diverse behavior, utilizing on average more than double the number of distinct routes (Mean = 10.91, SD = 10.90; Median = 7). It is particularly worth noting the high standard deviation within the e-cyclist group, which aligns with the long-tail distribution observed in the figure. This statistic confirms that there exists a subset of "high-explorers" who actively exploit over 20 different route alternatives.

Fig. 9 presents the distribution of the total number of valid trips for each respondent. A distinct distributional divergence is observed between the two modes. The distribution for traditional cyclists is heavily concentrated in the lower frequency range, with the majority (77.6%) of respondents contributing between 2 and 5 valid trips. In contrast, the e-cyclist group exhibits a more dispersed distribution characterized by a "long tail" of high-frequency usage. While 36.5% of e-cyclists fall within the [2, 5] range, a substantial proportion (38.6%) recorded more than 10 valid trips, compared to only 7.5% among traditional cyclists. Despite the variation in total trip volume, all respondents included in this analysis met the minimum data requirement (at least two valid trajectories). This ensures that every individual in the dataset—regardless of group—provided sufficient topological information to compute the VSRCB index.

Analysis of VSRCB

Comparative analysis of VSRCB

Fig. 10 illustrates the frequency distribution of the VSRCB index for both e-cyclists and traditional cyclists. A distinct distributional difference is observed between the two modes. The distribution for traditional cyclists is skewed towards the lower intervals (median = 0.284), with 67.2% of respondents recording values below 0.4, suggesting a tendency towards route repetition. Conversely, the e-cyclist distribution exhibits greater dispersion and a higher median value (median = 0.455), characterized by a distribution tail extending beyond 2.0. A Mann-Whitney U test confirms that this difference is statistically significant ($U = 29157.0$, $p < 0.001$), indicating that e-cyclists display a higher degree of variety seeking behavior compared to traditional cyclists.

To isolate the influence of transport mode from individual heterogeneity, a within-subject analysis was performed on a subset of "dual users" who utilized both electric and traditional bicycles during the observation period. Fig. 11 displays the paired VSRCB indices for these respondents, where the x-axis and y-axis correspond to values obtained during traditional cycling and e-cycling, respectively. The dashed diagonal line (y = x) demarcates the boundary of equivalence.

As illustrated in Fig. 11, a distributional asymmetry is evident, with a predominance of data points located in the upper-left region (shaded blue) above the diagonal, indicating that respondents exhibit higher variety seeking behavior when using e-bicycles. A Wilcoxon Signed-Rank Test was conducted to statistically validate this pattern. The analysis yielded a test statistic of 1276.0 ($p < 0.001$), statistically confirming that, within the same pool of individuals, e-cycling is associated with significantly higher VSRCB values than traditional cycling. This

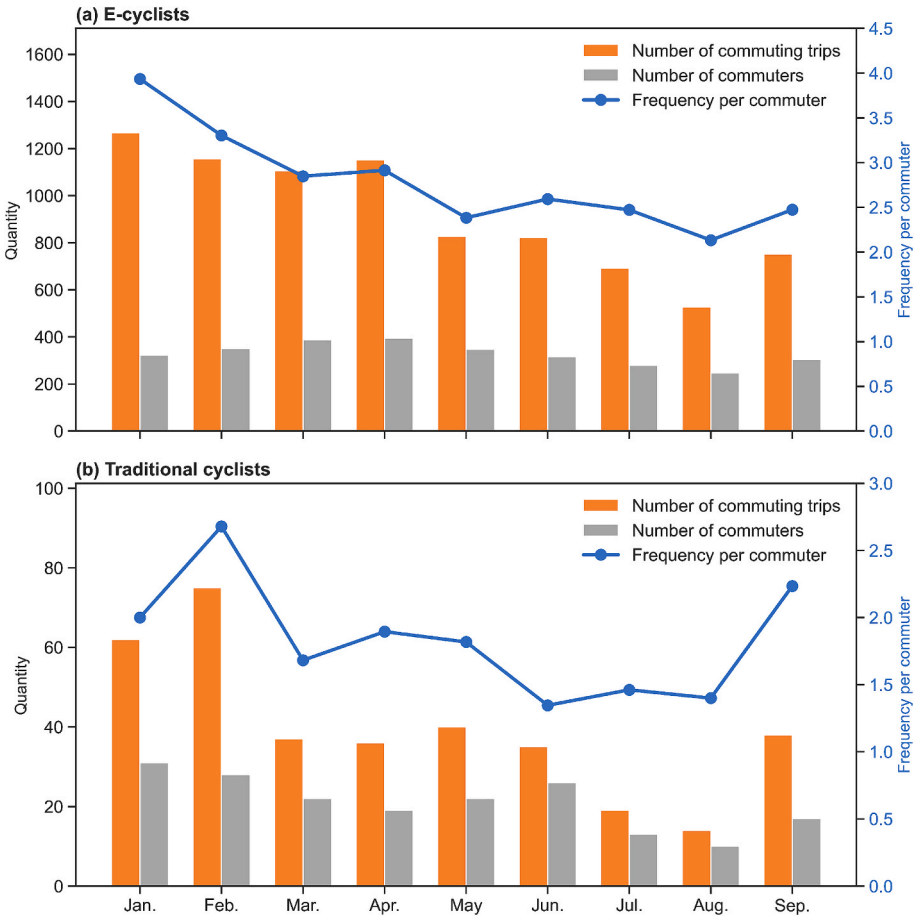


Fig. 7. Temporal variations in commuting volume and average frequency for (a) e-cyclists and (b) traditional cyclists.

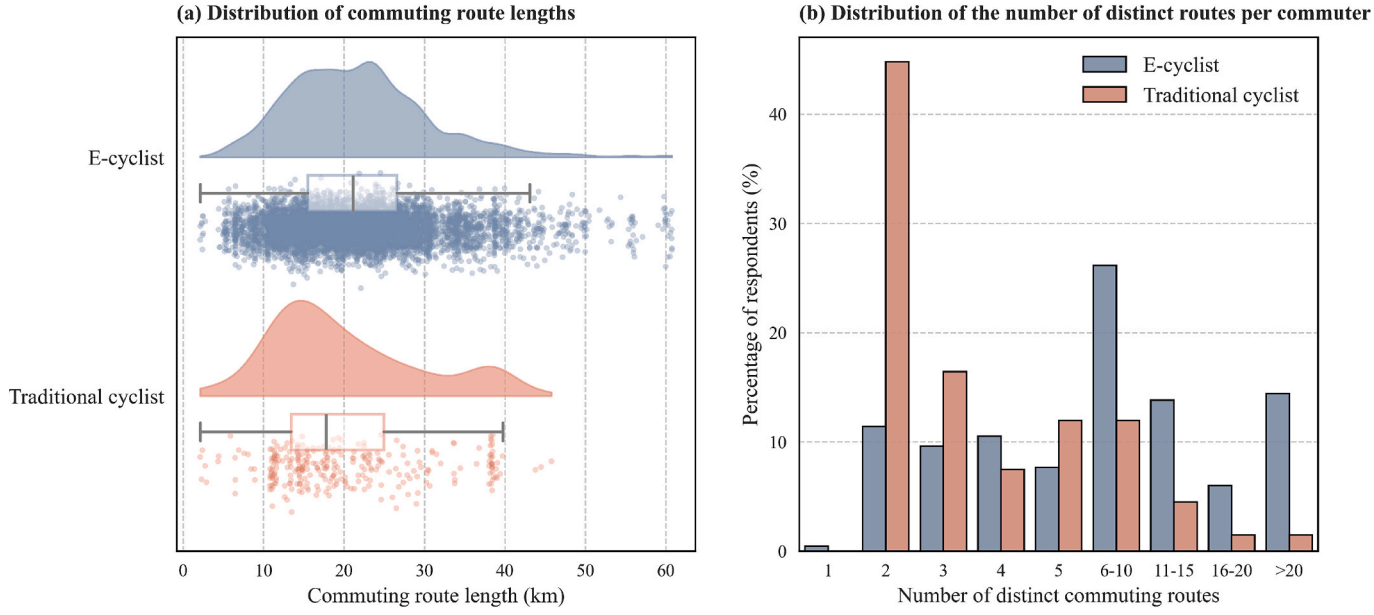


Fig. 8. Comparison of route characteristics between e-cyclists and cyclists.

finding implies that variety seeking behavior is moderated by the physical constraints of the transport mode. When operating a traditional bicycle, the requisite physical exertion imposes a high "impedance" to deviation, thereby constraining the rider's tendency to diverge from established paths. Consequently, the physiological cost of detouring likely suppresses the psychological desire for exploration, leading to route choices dominated by utility maximization. In contrast, the electric assist mechanism of e-bicycles effectively lowers the marginal physical cost of detouring, thereby "unlocking" latent variety seeking tendencies. By decoupling travel distance from physical exhaustion, e-

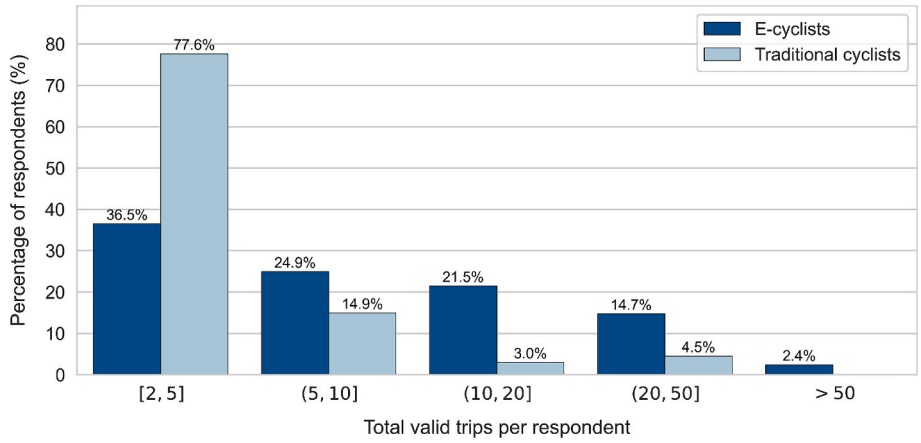


Fig. 9. Distribution of the total number of valid trips per respondent for e-cyclists and traditional cyclists.

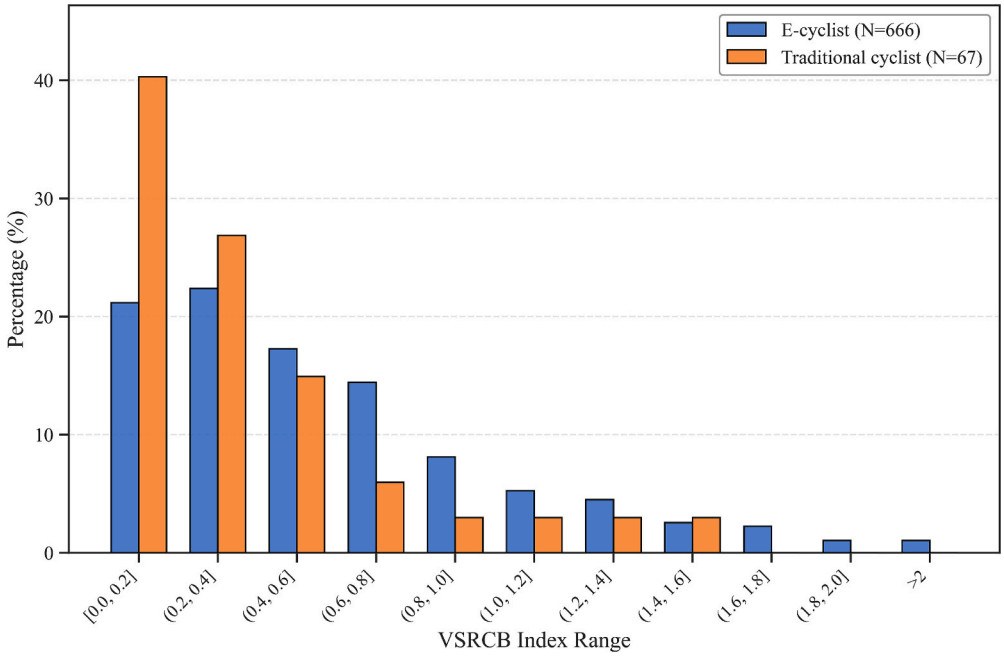


Fig. 10. Comparative frequency distribution of the VSRCB index for e-cyclists and traditional cyclists.

bicycles provide a form of "spatial emancipation," allowing intrinsic preferences—whether for scenery, safety, or novelty—to manifest more freely. Thus, the shift to e-mobility appears to reshape the spatial structure of commuting, transforming it from a strictly efficiency-oriented task into a more flexible activity.

Beyond the absolute increase in VSRCB levels, it is important to examine the stability of individual route choice behavior across transport modes. Specifically, we assess whether respondents who rank as high variety seekers relative to their peers when using an e-bicycle maintain a similar relative position when cycling with a traditional bicycle.

To this end, a Spearman's rank correlation analysis was conducted between individual VSRCB scores across the two modes. The results reveal a statistically significant but moderate-to-weak positive association ($\rho = 0.35, p = 0.010$). The statistical significance ($p < 0.05$) suggests that VSRCB is partly driven by intrinsic personal traits; some individuals are inherently more prone to exploration regardless of the vehicle. However, the relatively low correlation coefficient ($\rho = 0.35$) indicates that the "spatial emancipation" provided by e-bicycles does not affect all users uniformly. The shift in transport mode triggers a reshuffling of user rankings. This implies that the removal of physical constraints (as noted in the Wilcoxon results) interacts with individual preferences in complex ways, causing many users to drastically alter their relative exploration habits.

Decomposition of VSRCB dimensions

To elucidate the specific drivers underlying the observed variety seeking behavior, we further examine the three constituent dimensions of the VSRCB index outlined in Section 2.1: Quantity (Number of Routes), Balance (Shannon Entropy), and Distinction (Weighted Distinction Rate). Table 5 presents the comparative statistics for each dimension and VSRCB index. Additionally, the results of the Mann-Whitney U test are presented to assess the statistical significance of the differences between e-cyclists and traditional cyclists, where a p-value less than 0.05 indicates a statistically significant divergence in the distribution of the metric.

As indicated in Table 5, the decomposition analysis reveals that the significantly higher VSRCB observed in e-cyclists ($p < 0.001$) is not equally attributable to all dimensions; rather, it stems primarily from significant disparities in the Quantity and Balance dimensions.

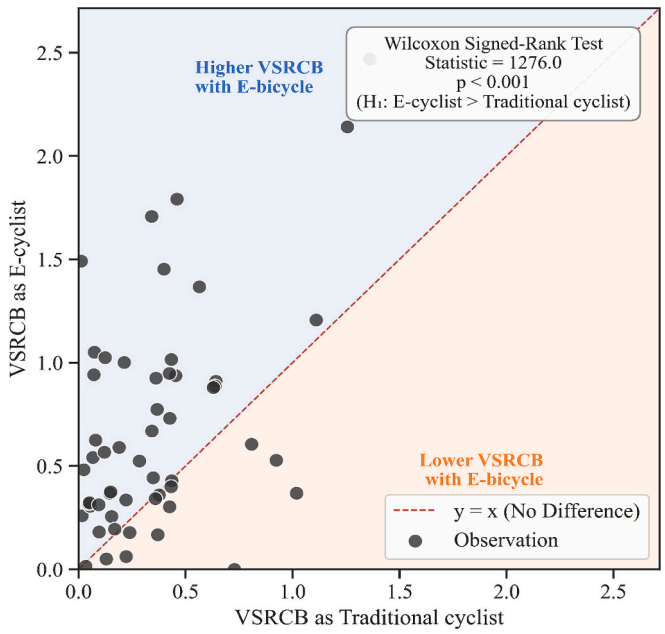


Fig. 11. Paired comparison of VSRCB values for the subset of dual-mode respondents.

Table 5
Descriptive statistics and comparative analysis of VSRCB dimensions for e-cyclists and traditional cyclists.

Dimension / Index	E-cyclists (N = 666) Mean ± SD	Traditional cyclist (N = 67) Mean ± SD	Mann-Whitney *U* TestP-value
Quantity	10.91 ± 10.90	4.48 ± 4.79	< 0.001 ***
Balance	1.97 ± 0.86	1.20 ± 0.61	< 0.001 ***
Distinction	0.20 ± 0.14	0.21 ± 0.18	0.700 (ns)
VSRCB Index	0.58 ± 0.47	0.37 ± 0.36	< 0.001 ***

Note: *** $p < 0.001$, ns = not significant.

First, the Quantity Dimension acts as a primary contributor. Consistent with the descriptive analysis, e-cyclists maintain a route set that is, on average, more than double the size of that of traditional cyclists (10.91 vs. 4.48). This confirms that the electric assist capability fundamentally expands the spatial consideration set of riders.

Second, the Balance Dimension (Shannon Entropy) shows a marked increase for e-cyclists (1.97 ± 0.86) compared to traditional cyclists (1.20 ± 0.61). This is a critical finding: it indicates that e-cyclists do not merely accumulate more "backup" routes; they actively utilize their route options with greater regularity. Unlike traditional cyclists, who tend to have a dominant route preference (low entropy), e-cyclists exhibit a more flexible switching behavior, distributing their trips more evenly across their expanded set of alternatives.

However, the Distinction Dimension shows no statistically significant difference between the groups (0.20 vs. 0.21, $p = 0.700$). The Distinction Rate measures the geometric uniqueness of the chosen routes (i.e., how much the routes overlap spatially). The low and similar values in both groups suggest that, regardless of the transport mode, the physical constraint of the road network and the fixed OD pairs limit the extent to which completely disjoint routes can be chosen. Most alternative routes inevitably share certain segments (e.g., main arterial roads).

In summary, the superior variety seeking behavior of e-cyclists is driven by a "Quantity + Balance" mechanism. E-bicycles empower users to discover more path options and switch between them more frequently, even if the geometric uniqueness of these paths remains constrained by the urban grid structure similar to traditional cycling.

Finally, to determine the relative contribution of each component to the VSRCB index, we analyzed the Spearman correlation coefficients between the composite index and its three constituent dimensions. The results are presented in Table 6.

As shown in Table 6, all three dimensions exhibit positive and statistically significant correlations with the composite VSRCB index ($p < 0.001$) across both e-cyclist and traditional cyclist groups. However, the strength of these associations varies.

The Quantity Dimension and Balance Dimension show moderate correlations with the VSRCB index (ranging from $\rho = 0.561$ to 0.637). This suggests that while the number of routes and the evenness of their usage are foundational elements of variety seeking, they do not individually account for the full variability of the composite metric.

In contrast, the Distinction Dimension demonstrates the strongest correlation with the VSRCB index (E-cyclists: $\rho = 0.849$; Cyclists: $\rho = 0.877$). This strong association indicates that the VSRCB index is particularly sensitive to the geometric uniqueness of the chosen paths. Consequently, the metric reflects a weighted integration where spatial distinctiveness plays a predominant role alongside route quantity and usage balance.

VSRCB variation across Socio-Demographic profiles

To determine whether the observed variety seeking behavior is associated with the cyclists' socio-demographic backgrounds or the spatial characteristics of their home and work locations, we examined the distribution of VSRCB values across gender, age groups, and urbanization levels. Table 7 summarizes the descriptive statistics and the results of non-parametric statistical tests.

As presented in Table 7, the VSRCB for e-cyclists appears largely unrelated to socio-demographic profiles. Statistical analysis revealed no significant disparities across gender ($p = 0.241$) or age groups ($p = 0.815$). This suggests that the propensity for route exploration using e-bicycles is not confined to specific demographic subsets but is rather a pervasive behavior across the user base.

Conversely, the level of urbanization appears to be a relevant factor for VSRCB ($p = 0.004$). Respondents residing in highly urbanized areas (Level I) exhibited the highest variety seeking tendency (mean = 0.74), compared with those in moderately (Level II, mean = 0.56) or sparsely (Level III, mean = 0.55) urbanized regions. This disparity likely reflects the structural characteristics of the road network: highly urbanized environments typically feature dense, grid-like networks offering multiple routing alternatives, whereas suburban or rural settings (Level III) often rely on limited arterial roads, which physically constrain route choice options.

The analysis of home-work spatial combinations lends further support to this infrastructure-driven hypothesis. Although the overall variation was only marginally significant ($p = 0.058$), the data indicate that commutes originating and terminating in highly urbanized areas (e.g., Pair I-I, mean = 0.77; Pair I-II, mean = 0.78) tend to yield higher VSRCB values compared to those within less urbanized settings (e.g., Pair III-III, mean = 0.54).

Regarding traditional cyclists, no statistically significant associations were identified across any socio-demographic or spatial categories (all $p > 0.05$). VSRCB values for this group remained consistently lower, irrespective of age or location. In summary, these findings suggest that

Table 6
Spearman correlation coefficients between the composite VSRCB index and its constituent dimensions for e-cyclists and traditional cyclists.

Dimension	E-cyclists (N = 666)		Traditional cyclist (N = 67)	
	Coeff.	P-value	Coeff.	P-value
Quantity	0.618	< 0.001 ***	0.561	< 0.001 ***
Balance	0.637	< 0.001 ***	0.574	< 0.001 ***
Distinction	0.849	< 0.001 ***	0.877	< 0.001 ***

Note: *** $p < 0.001$.

Table 7
Descriptive statistics of VSRCB across socio-demographic categories for e-cyclists and traditional cyclists.

Factor	Category	E-cyclist (N = 666)			Traditional cyclist (N = 67)		
		N	Mean ± SD	P-value	N	Mean ± SD	P-value
Gender	Female	338	0.55 ± 0.42	0.241	30	0.35 ± 0.32	0.880
	Male	328	0.62 ± 0.52		37	0.39 ± 0.40	
Age Group	≤ 45	158	0.63 ± 0.57	0.815	16	0.43 ± 0.43	0.203
	(45,55]	315	0.55 ± 0.42		25	0.26 ± 0.20	
	> 55	193	0.59 ± 0.46		26	0.45 ± 0.42	
Home Urbanity	I	113	0.74 ± 0.58	0.004 **	12	0.37 ± 0.34	0.274
	II	139	0.56 ± 0.44		12	0.57 ± 0.50	
	III	414	0.55 ± 0.44		43	0.31 ± 0.31	
Work Urbanity	I	295	0.60 ± 0.46	0.357	33	0.38 ± 0.37	0.356
	II	106	0.60 ± 0.50		7	0.31 ± 0.48	
	III	265	0.56 ± 0.47		27	0.38 ± 0.34	
Home-Work Pair	I-I	54	0.77 ± 0.52	0.058 †	8	0.33 ± 0.22	0.613
	I-II	24	0.78 ± 0.53		–-	–-	
	I-III	35	0.67 ± 0.69		–-	–-	
	II-I	72	0.58 ± 0.46		6	0.64 ± 0.57	
	II-II	24	0.53 ± 0.40		–-	–-	
	II-III	43	0.53 ± 0.44		6	0.51 ± 0.45	
	III-I	169	0.56 ± 0.43		19	0.31 ± 0.31	
	III-II	58	0.55 ± 0.51		5	0.35 ± 0.57	
	III-III	187	0.54 ± 0.43		19	0.31 ± 0.23	

Note: Statistical significance was assessed using the Mann-Whitney *U* test for two groups (Gender) and the Kruskal-Wallis H test for three or more groups. $^{\dagger}p < 0.10$, $^{**}p < 0.01$. Groups with a sample size of fewer than 5 respondents were excluded from the analysis to ensure statistical reliability.

variety seeking behavior in route choice may be less dependent on the cyclist's demographic profile (age, gender) and more closely aligned with the spatial context (network density) and the specific mobility tool employed.

Conclusion

This study investigates the route choice behaviors of e-cyclists through the lens of VSRCB. Using a large-scale GPS dataset from the Netherlands and a multi-dimensional measurement framework, we quantify the extent to which e-cyclists deviate from habitual routes and compare these patterns with those of traditional cyclists. The empirical analysis yields several insights into the relationship between transport mode and route choice behavior.

First, the results confirm that VSRCB is an observable behavioral phenomenon present in both traditional and electric cycling, indicating that the intrinsic motivation to vary routes exists regardless of the mode. However, the empirical evidence demonstrates that the transition to electric cycling is associated with significantly higher levels of this variety. E-cyclists exhibit a VSRCB index that is statistically higher than that of traditional cyclists. The within-subject analysis of dual-mode users serves as a pivotal finding, demonstrating that the same individual displays significantly greater route diversity when riding an e-bicycle compared to a traditional bicycle. This finding supports the notion of spatial emancipation, whereby electric assistance reduces the physical constraints associated with distance and effort. As a result, constraints that typically limit exploratory route choice in conventional cycling are relaxed, allowing greater expression of individual route choice flexibility. The observed increase in variety seeking suggests a shift away from strictly distance- or effort-minimizing behavior toward a broader utility-based decision framework in which the marginal cost of detours is substantially reduced.

Second, the decomposition of the VSRCB index provides insight into the mechanisms through which route choice variety increases. The higher VSRCB values observed among e-cyclists are primarily driven by the Quantity (number of unique routes) and Balance (entropy of route usage) dimensions, rather than by Distinction (geometric uniqueness). This indicates that e-cyclists do not necessarily discover spatially disjoint shortcut or scenic paths that are completely separate from the main network; rather, they utilize the existing grid more extensively, switching frequently between a larger set of available parallel options. The absence of a significant difference in the Distinction dimension suggests that, for both transport modes, route geometry is largely shaped by the underlying road network topology and the fixed origin–destination configurations, which impose structural constraints on the degree of attainable spatial divergence.

Third, the analysis explores the association between VSRCB and cyclist profiles. The results indicate no statistically significant differences in VSRCB across gender or age groups, suggesting that route choice variety does not vary with these demographic characteristics. With respect to spatial context, a statistically significant difference in VSRCB distributions is observed across levels of residential urbanization within the e-cyclist group, whereas no comparable association is found among traditional cyclists. For e-cyclists, those residing in highly urbanized areas (Level I) exhibited higher average VSRCB values compared to those in less urbanized regions (Levels II and III). This mode-specific contrast points to an interaction between physical capability and spatial opportunity in shaping variety seeking behavior. Highly urbanized environments provide dense and well-connected road networks that facilitate route variation; however, this potential appears to be more fully realized by e-cyclists, whose physical effort constraints are reduced by electric assistance. Traditional cyclists, conversely, appear limited by physical constraints regardless of the spatial context, preventing them from exploiting the route diversity even in high-density areas.

These findings provide several planning-relevant implications for urban cycling infrastructure design. First, regarding investment decisions, the observed tendency of e-cyclists to actively utilize multiple route alternatives suggests that cycling network planning may benefit from moving beyond an exclusive emphasis on single, high-speed corridors such as cycle superhighways. Instead, greater attention could be directed toward enhancing network redundancy by ensuring the availability of parallel and well-connected secondary routes. Such a mesh-like network structure is better aligned with observed route-switching behavior and may enhance the overall resilience of the cycling system. Second, the willingness of e-cyclists to accept moderate detours indicates a potential for more balanced spatial distribution of cycling flows across the network. This behavioral flexibility indicates that promoting e-bicycle adoption can serve as a strategic tool to alleviate peak-hour congestion, as e-cyclists are less bound to single efficiency-maximizing corridors. Furthermore, this flexibility can be leveraged through soft policy measures, such as dynamic wayfinding systems, to actively redirect cyclists. Since electric assistance reduces the marginal effort of detouring, such promotional and management strategies are highly feasible for optimizing urban network capacity. Finally, the lower VSRCB observed in less urbanized areas points to a potential limitation in the availability of viable route alternatives. To promote active

mobility in suburban and *peri*-urban contexts, infrastructure investments could prioritize the creation of loop connections, local shortcuts, or alternative paths that reduce reliance on single arterial routes. By expanding the local choice set, such interventions may improve the overall attractiveness of cycling in lower-density environments.

This study acknowledges several limitations. First, the dataset employed is derived from 2014. While the fundamental bio-mechanical advantage of e-assistance remains constant, the cycling infrastructure landscape may have evolved since the data collection period. Second, although this study controlled for fixed spatial anchors (home and work locations), it did not account for dynamic environmental covariates—such as meteorological conditions or temporary network disruptions (e.g., road construction)—which may necessitate transient route deviations. Future research should prioritize the integration of longitudinal data to disentangle "voluntary" variety seeking (driven by intrinsic motivations like curiosity) from "forced" deviations (imposed by extrinsic constraints like congestion), thereby providing a more nuanced understanding of the behavioral mechanisms underlying e-cycling.

CRediT authorship contribution statement

Xueting Zhao: Writing – original draft, Software, Methodology, Formal analysis. **Peter J.H.J. van der Waerden:** Writing – review & editing, Investigation, Methodology, Resources, Supervision, Data curation, Conceptualization. **Peng Jia:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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